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Keystone management species under the Endangered Species Act can promote ecosystem-based management

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Human activities such as land conversion and resource consumption can cause species declines, which in the United States can result in listing for protection and recovery under the Endangered Species Act. Often highly charismatic species, such as bald eagles, grizzly bears, Pacific salmon, and North Atlantic right whales, get linked to human activity and social conflict emerges. This conflict can drive significant investment in a species' recovery, including the suspension or alteration of multiple human activities, particularly when the species' needs overlap with human activities such as food production, water management, and energy extraction. Analogous to keystone species, which directly influence their ecosystem through biological activity, we define these endangered and threatened species as keystone management species (KMS), because they prompt people to alter or undo their impacts on ecosystems through management activities that might not occur without that species' charisma or social value. The phenomenon of these KMS under the ESA appears to be single species management at a surface level, but it is much more complex, often with cascading benefits to other species, while also increasing coordination across multiple sectors, resulting in ecosystem-based management. We review several case studies including the bald eagle's emergence as one of the first KMS, following its dramatic recovery after the banning of DDT and widespread reintroduction projects; Pacific salmon's influence on western land and water management and hydropower; the grizzly bear's influence on western land and community waste management; and the North Atlantic right whale's current influence on commercial fisheries and shipping and emerging influence on offshore aquaculture, marine renewable energy, and other marine activities.

KEYWORDS

endangered species act, keystone-management species, ecosystem-based management, endangered species investments, charismatic species, flagship species, umbrella species

1 Introduction

1.1 The ESA and EBM

In December 1973, the United States Congress passed the Endangered Species Act (ESA), a groundbreaking piece of conservation legislation, thus beginning a significant social experiment. More than 50 years later, the ESA remains a powerful and quite controversial piece of legislation as it comes into increasing conflict with human activities, such as agriculture, fisheries, and energy development, and impacts such as habitat degradation and climate change. The human population has doubled from 4 billion to 8 billion during this period. People and livestock now make up about 96% of global mammalian biomass (Bar-On et al., 2018). Climate change is shifting ecosystems at rates faster than our worst-case models anticipated just a few years ago (IPCC, 2022; Tollefson, 2022). As a result, we are observing the highest rates of extinction in roughly 65 million years and have likely prompted Earth's sixth mass extinction event (Barnosky et al., 2011).

These staggering statistics provide background context to the effectiveness of the ESA in the United States. While criticized for rarely achieving recovery once a species is listed, the ESA has maintained survival for more than 95% of the species protected by the act (Evans et al., 2016). And yet its original approach, focusing on protecting individual species, has come to be thought of as outdated with its focus on single species management and recovery.

During the 1990s, marine resource managers recognized the challenge of single species management practices and proposed new ways of thinking about managing fisheries within the construct of Ecosystem-Based Fisheries Management, which NOAA's National Marine Fisheries Service (NMFS) defines as "a systematic approach to fisheries management in a geographically specified area that: contributes to the resilience and sustainability of the ecosystem; recognizes the physical, biological, economic, and social interactions among the affected fishery-related components of the ecosystem, including humans; and seeks to optimize benefits among a diverse set of societal goals" (Link, 2024). This was a significant step in recognizing direct and indirect ecosystem pressures and the consequences of managing stocks in isolation. Yet EBFM remains reductionist, approaching ocean management from a single use: fishing. The concept of Ecosystem Based Management (EBM) goes beyond fisheries to recognize a more holistic view of natural resources and threats across marine and terrestrial ecosystems. Thirty years into EBFM/EBM theory we are still struggling to transition management practices toward those approaches (Haugen et al., 2024). There are probably many reasons for this, but among them are concerns that EBM may be too complex for the average citizen or even scientist to comprehend or put into practice. Indeed, the irony recognized by Leopold (1966) and others since, is that the average citizen thinks "scientists" can do this, while practical ecologists and managers realize ecosystems are too complex to fully quantify and manage precisely. Recovery goals based on stable ecosystems are increasingly challenged by shifting baselines driven by forces beyond US regulatory control (e.g.

human population growth, climate change). As we have endeavored to manage biodiversity and natural resources, there has been an emphasis on high profile species that have a large economic value or are charismatic and valued for other reasons (Evans et al., 2016; Gerber, 2016; Haines et al., 2021). Resulting litigation and regulatory actions can shape the efforts of management agencies, pressuring them to divert significant portions of their resources into a few species, resulting in a framework that resembles single species management.

The conservation literature includes extensive debate over the relative merits and tradeoffs between how the focus on a few key species have cascading benefits to many other species and ecosystem preservation versus how a single species focus distracts from the needs of other less charismatic species and the real problems of human population growth, habitat loss, and climate change. The literature is peppered with terms that touch upon this single species focus including "charismatic megafauna" and "celebrity species"; the terms "umbrella species" and "flagship species" tend to be more positive, acknowledging extended benefits to other species and ecosystems (Andelman and Fagan, 2000; Jarić et al., 2024; Runge et al., 2019; Thompson, 2010). The clear disparity in investment between a few charismatic and/or economically valuable species and the minimal investment for most protected species has led to concerns about funding allocation. Indeed, we compared investment across taxonomic kingdoms from recent United States Fish and Wildlife Service (USFWS) reports (2017-2020): vertebrates receive 94% of all funding but represent only 31% of all listings, with 5 species of Pacific salmon receiving 44% of all investment during that period.

Ardiantiono et al. (2024) compared the extended benefits of monitoring umbrella species selected through popular charisma versus applying a criterion for umbrella taxa that represented local biodiversity patterns and found that charismatic species served as poor indicators in some cases, implying different management approaches might benefit biodiversity. It is worth noting that over the course of 50 years, while the USFWS and NMFS (or for both agencies: the Services) retain technical authority to prioritize recovery actions across endangered species, most recent funding (~74%) comes from other federal and state agencies (United States Fish and Wildlife Service (USFWS), 2019). And often, new congressional appropriations to the Services prescribe explicit investment in a small number of high-profile species typically in response to stakeholder litigation, leaving the Services with limited discretionary resources to promote specific recovery actions for species that lack stakeholder conflict.

1.2 Keystone management and species

Below we describe the emergence of an analytical concept, in which single-species management to recover a few listed species under the ESA drives ecosystem-based management. The specific circumstances when this occurs typically includes the species being charismatic, having additional protections under other legislation, and at some point being highly endangered with recovery needs that

are in conflict with fundamental human needs such as safety, water, food and energy production. Under these circumstances, some endangered species protection and recovery efforts move beyond the standard practices used for many umbrella species, where the primary tool is habitat preservation, with benefits extending to other species. This practice of drawing a box and saying ‘they can live here and we won’t bother them’ is of decreasing value given the ubiquity of humans across the landscape (Greenspoon et al., 2023). When a species’ survival becomes interwoven with human activities, and sparks a change in management, that species’ influence becomes keystone in nature.

In ecology, the term keystone species is used when a single species has a disproportionate impact on its environment relative to its abundance, thus driving the distribution and abundance of other species in its ecosystem (Paine, 1969). A classic example is the role sea otters (*Enhydra lutris*) play in establishment of kelp forests along the Pacific Coast of North America (Estes and Palmisano, 1974). Our observation is that some endangered species protected under the ESA reach a point where management decisions become high profile with large management costs and ecological impacts because a species’ recovery needs are in direct conflict with fundamental human activities. When the necessary actions to recover these species subsequently drives multi sector coordination around its requirements, this approaches EBM. We propose the term “keystone management species” (KMS) for this phenomena (Figure 1).

Keystone management species can affect both natural and developed environments. Specifically, humans act to preserve or restore ecosystem elements required by the KMS and to curb anthropogenic activities that endanger them, typically with considerable economic investment. While often a single species focus, these changes affect many other organisms in an ecosystem (the umbrella/flagship species component). The reason for distinguishing KMS from umbrella/flagship phenomena is that these species, often having large range requirements, cannot be separated from humans, and their requirements are in direct conflict with economic activities or fundamental resource needs

such as food, water or energy security. Unlike an ecological keystone species and perhaps in contrast, the species’ effect on the ecosystem (due to human intervention) can increase as the species becomes more rare, having minimal ecological effect on its own.

There are significant implications for this concept in US policy and abroad. In general, this paper stops short of policy recommendations, seeking primarily to increase awareness of the phenomenon and to inform those engaged with driving policy and decision making.

1.3 Endangered species investments

This manuscript was inspired by observations across decades of working on protected species. We have heard many concerns voiced by conservationists and academics about investments in charismatic species over less celebrated ones. Similar debates occur in resource-management agencies. Why is funding orders of magnitude higher for Pacific salmon when people eat groundfish, too? And why are stakeholder debates around some species, such as North Atlantic right whales (*Eubalaena glacialis*), amplified to greater levels than other species at greater risk of extinction like the North Pacific right whale (*Eubalaena japonica*)? These thoughts led us to look at government investment across species.

Data quantifying investments are available in a series of annual reports from the USFWS as mandated by Congress in the 1988 reauthorization of the ESA (USFWS, 1990–2021). For each report, the USFWS requests accounting by all relevant federal and state agencies for investments in ESA listed species. These reports constitute a good faith effort to account for annual ESA investments, though they caution against assumed precision due to interannual differences in how expenditures are calculated and reported (or not) by federal and state agencies. Most notable of these uncertainties is the 2019 report that suggests a 50% drop in ESA investments from 2018. Rather the drop is likely related to being generated during the peak of the Covid 19 epidemic when 26 of 34 federal agencies did not provide annual cost estimates.

For the purposes of this paper’s discussion, these reports capture how the vast majority of investment goes to a small fraction of all listed species. Pacific salmon, for example, which are represented by 28 stocks across 5 species, received 43% of all reported investment across 1912 listed species in 2018 (United States Fish and Wildlife Service (USFWS), 2019). It is worth acknowledging the range here: there were years when single stocks of Pacific salmon received investments in excess of \$100 million per year while other listed species saw almost zero dollars.

All estimates are drawn from USFWS report Tables 1 (activity related expenditures) and 3 (land acquisitions) that could be tracked to the individual species or stock level. In cases where multiple lines of funding were reported to a species or stock (e.g., separated by experimental versus endangered populations), all funding for that species/stock was summed. Reports were available through 2020 at the time of writing in 2025. As NMFS is the primary entity responsible for reporting all investments in North Atlantic Right Whales (*Eubalaena glacialis*), the values for accounting provided to



FIGURE 1
Keystone management species have the potential to affect management decisions of human activities and natural habitats thus driving ecosystem-based management.

USFWS for future reports (2021-2025) were available to the lead author and included in Figure 2. A more detailed description of the reports and how they were used is included in appendix A.

2 Criteria and case studies

The status of being a KMS is likely dynamic, emerging when the conditions are right and perhaps declining after successful recovery. We chose three species and one genus (Pacific salmon, Figures 1, 2) to explore this concept of KMS and potential influence in EBM. A common question in early discussions and reviews of this paper is why these four? There were a range of reasons, some as simple as personal and national familiarity, but the five criteria that seem to best define the threshold for when an endangered species becomes a KMS are:

1. Broad Range/Infeasible Separation: The species' extensive habitat range makes protection through wildlife reserves impractical due to the massive scale; separating the species from human activity is not feasible.
2. Direct Conflict with Essential Human Activity: A highly valuable or fundamental human activity that causes unsustainable mortality is directly preventing recovery of a species with a high probability of extinction.
3. Regulatory/Resource Trigger: Conflict necessitates regulatory action with economic consequences for stakeholders or limits human resource availability.
4. Directed, Sustained Investment: Explicit legislative or agency funding is allocated to resolve the conflict, ensuring both species recovery and human needs are met through a consistent, high, or increasing level of multi-year investment (minimum \$5M/year in current US dollars for

examples below) typically focused on modifying human activity.

5. Ecosystem-Based Influence: Management actions for the species' recovery generate clear, positive cascading effects, benefiting other species across the ecosystem.

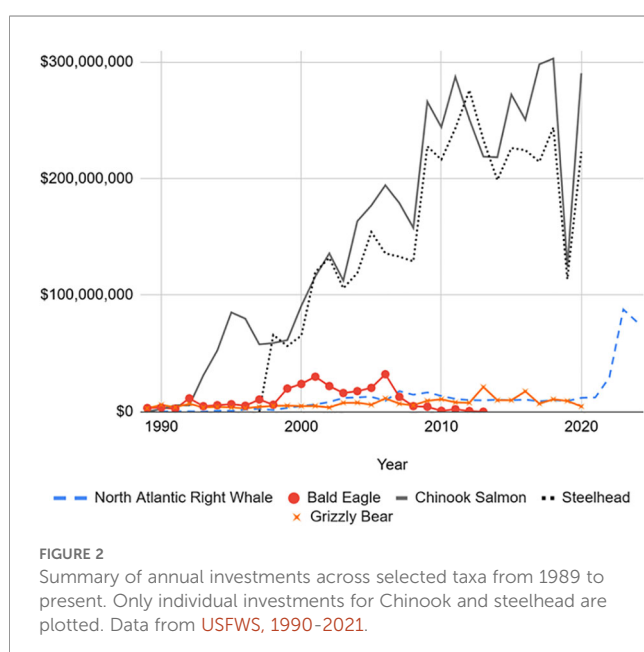
We acknowledge that several other species likely qualify and that not all KMS species will meet all criteria".

2.1 Bald eagle *Haliaeetus leucocephalus*

After the dangers of dichlorodiphenyltrichloroethane (DDT) were publicized by Rachel Carson's *Silent Spring* (1962), it was subsequently banned by the EPA for most uses in 1972 (United States Environmental Protection Agency, 1975; Grier, 1982). Carson described its widespread use and subsequent presence in our crops, dairy products, fish, rivers, and lakes, highlighting the impacts on wildlife, including songbirds, salmon, and bald eagles. While only a brief part of her thesis, she documented reports of the eagle's decline in the 1940s and 50s, commenting that such a trend 'may well make it necessary for us to find a new national emblem'.

Carson's work on DDT in many ways gave birth to the modern environmental movement with a broad review of ecosystem impacts. Then a critical next step occurred. A small group of concerned scientists, birders and lawyers formed the Environmental Defense Fund and for the very first time a citizen group brought suit to the government for an environmental issue (Wurster, 2015) giving birth to the concept of conservation litigation in the emerging field of environmental law. Wurster and colleagues built their list of DDT concerns around a portfolio of ecological and taxonomic concerns from Pacific salmon (in the Great Lakes) to raptors with emphasis on peregrine falcons (*Falco peregrinus*), osprey (*Pandion haliaetus*) and bald eagles. This began with a series of regional suits in 1966, then in 1970 reached a national level with the U.S. Court of Appeals for the District of Columbia Circuit. Along the way concerns about raptors and eggshell thinning became part of their case. And while Wurster comments that perhaps more focus was on peregrines at the time due to their popularity with falconers, a *Science* reporter published a story on the case (Carter, 1969) and the journal put a photo of a bald eagle chick on its cover, foreshadowing the species future prominence.

It seems that bald eagles were not the key driver in the initial movement to ban DDT. Rather it was broad public concern around a range of ecological impacts and emerging health issues. But this species has emerged as the greatest success story of this foundational moment in conservation history, justifying the continued need for such action and concern. The bald eagle was listed in 1967 under the Endangered Species Preservation Act, after only 417 breeding pairs were observed in 1963, however it was not until after DDT was banned in 1972 that this eagle was again listed in the 48 contiguous states under the ESA in 1978. The eagle's subsequent and surprising recovery was associated with reduced concentration of DDT (specifically DDE) in eggs and widespread



restoration and release of bald eagles in historical breeding grounds (Grier, 1982). A recent assessment reports more than 71,000 breeding pairs and 316,700 individuals (United States Fish and Wildlife Service (USFWS), 2020a). The bald eagle was removed from the ESA in 2007 but remains protected by the Migratory Bird Treaty Act and the Bald and Golden Eagle Protection Act.

Today, the bald eagle is an emblem of the environmental movement. The magnitude of its recovery post DDT seems to have played a huge role in amplifying the concerns around the example of DDT, that chemicals can be a significant health hazard to humans and the environment. It was interesting that the movement to ban DDT really began as a concern over wildlife declines, with linkages to human health just starting to emerge after the movement was underway (Wurster, 2015) when DDT was banned in 1972. It was much later that the full suite of human health hazards was realized (Rogan and Chen, 2005) and the chemical was banned (for most uses) more globally by the Stockholm Convention on Persistent Organic Pollutants in 2004 in response to both human health and environmental concerns.

Data on early US investments in bald eagle recovery are difficult to come by before 1989, making the magnitude of early investments uncertain. However it was a 'top 10' species for highest investment for many years after that. It peaked from 1999 to 2006, averaging \$22 million per year, before the 2007 delisting and subsequent end of dedicated ESA funding in 2012 (Figure 2). It's uncertain if this species achieved KMS status during the early litigation phase to ban DDT but it's clear that ending the use of DDT in this country had dramatic broad ranging effects and the eagle emerged as one of the most showcased. It was likely during this recovery period that it really emerged as a KMS and their influence drove critical habitat protections around nesting areas and efforts to minimize land development and human disturbance all due to their social status. Many of those protections remain in place under the Bald and Golden Eagle Protection Act and Migratory Bird Treaty Act affecting a range of human activities including forestry, construction, agriculture, recreational boating, and off-road vehicle and aircraft use (USFWS, 2007b). The bald eagle is likely to continue its role as a keystone management species, in part because of its continued protections under other laws, but also because of its iconic position in the United States, as we continue to try to establish a balance between human activities and the protection of wild animals and their habitats.

2.2 Pacific salmon/steelhead genus *Oncorhynchus*

"The Pacific Northwest is simply this: wherever the salmon can get to. Rivers without salmon have lost the life source of the area."
— Timothy Egan, 2011.

Hatched from eggs in gravel beds of rivers and streams, young salmon swim to sea and return just a few years later as large silver fish only to transform again. Their scales tighten into the body, a stunningly colored armor against the challenges of ascending rapids, battles between males and females digging to displace tons

of gravel to lay their eggs, and ultimately die (exception to steelhead), their carcasses feeding their offspring and surrounding land as well. Human use of coastal watersheds and oceans impacts them at every life stage.

In many, if not most endangered species conflicts, a given species often has a polarizing effect between consumers and conservationists. If there is a taxonomic group that crosses this divide, it is Pacific salmon. Whether your concern is about access to wild caught fish at restaurants or home, or to watch them spawn in high mountain streams (and often both), their life history is well known and much of the public wants there to be more salmon. It's this strong public sentiment that has driven the unprecedented investment in their recovery starting in the 1990s after what has been an unfortunate and long period of decline.

Salmon populations or stocks are managed regionally as distinct population segments or evolutionary significant units. Pacific salmon ESA listings began with the Sacramento River Winter-run stock of Chinook salmon (*Oncorhynchus tshawytscha*) in 1990 (Federal Register, 1990). This was followed by Snake River listings for Spring/Summer and Fall run Chinook and sockeye (*O. nerka*) a few years later. By 2001 there were listings for 9 stocks of Chinook, 2 stocks of Chum (*O. keta*), 2 stocks of coho (*O. kisutch*), and 10 stocks of steelhead (*O. mykiss*). Today NMFS manages listings for a total of 28 stocks of Pacific salmon with additional candidates and petitions under review.

The primary threats to these species are commonly noted by the four H's: Hydropower, Habitat Loss, Harvest and Hatcheries (National Research Council, 1996, National Research Council, 2004). Although technically within the context of habitat and hydropower, a fifth H, H₂O, has emerged as a limiting factor for many southern populations. Basically, fish are competing for access to freshwater water flows that are being altered and reduced by water flow regulation through dams and diversions for agricultural and municipalities. And climate change is exacerbating all of these threats (Kocik et al., 2022).

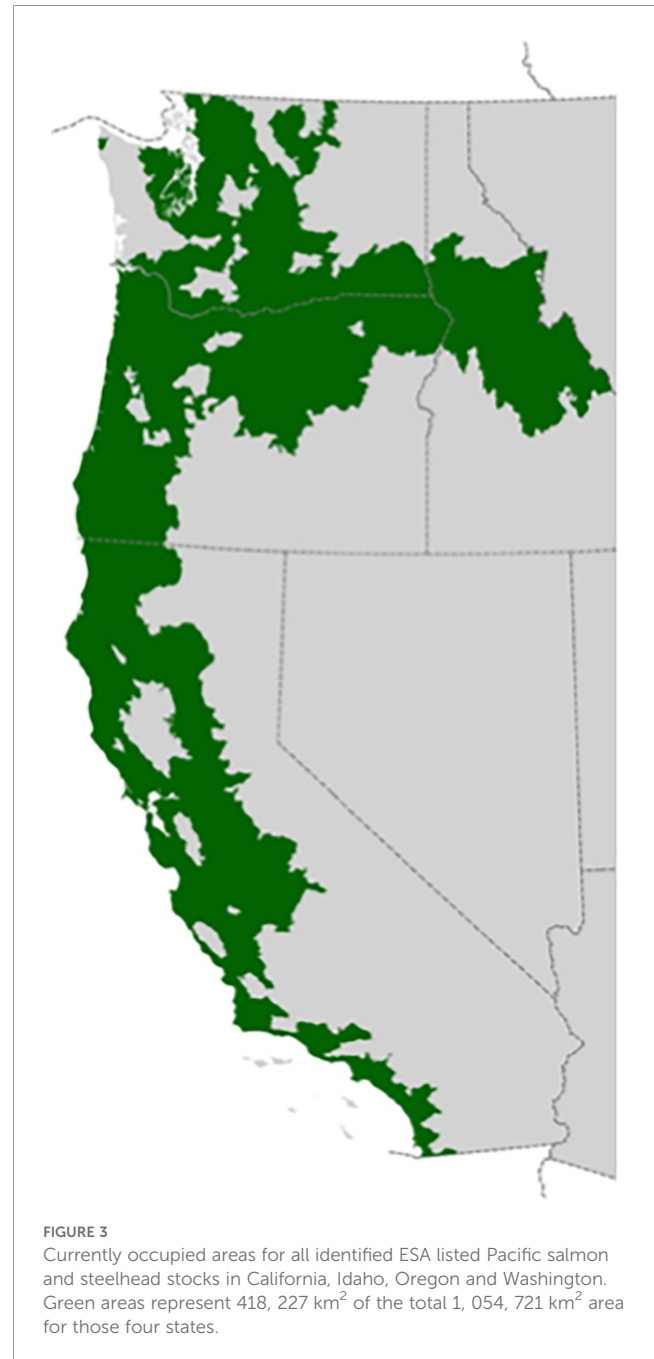
Pacific and Atlantic salmon (*Salmo salar*) have characteristics of keystone management species. Both groups are frequently identified as ecological keystone species (e.g. Helfield and Naiman, 2006; Woodward et al., 2021). But they also have unprecedented management influence. Atlantic salmon management follows many of the guiding principles of EBFM aside from lack of harvest (Hare et al., 2019). These EBFM principles especially apply to Pacific salmon management given there is often significant co-management of harvestable stocks in large river basins. Kocik et al. (2022) identified Pacific salmon as having keystone management impacts given the social power around salmon and the way their life history links many ecosystems from the Pacific high seas to mountain forests. These stressors are often dealt with independently and regionally, unsurprising given the sheer amount of investment and underlying stock complexities. Yet, these linkages require some level of functional habitat from the forests to the sea, integrating watershed-ocean management through the lens of Pacific salmon recovery.

When one considers management of ESA listed salmon stocks in the four western states of California, Idaho, Oregon, and

Washington, the social, political, and ecological influence of these species in the United States is taken to an unrivaled economic level, receiving roughly 40% of all investments toward ESA recovery for 5 species of salmon across 28 stocks (USFWS, 1995–2020). It is worth noting that the majority of these investments are not provided by the regulatory services (NMFS and USFWS) but other state and federal agency funding and ratepayer fees received by those agencies responsible for providing basic utilities such as hydropower and water (e.g. Bonneville Power Administration, the California Department of Water Resources, the Army Corps of Engineers, and the Bureau of Reclamation). In the Columbia River watershed, Bonneville Power has invested tens of billions of dollars on the recovery of multiple stocks of Pacific salmon to ensure the coexistence between both protected and harvested stocks of salmon and steelhead while providing much of the electricity to the Pacific Northwest along with multiagency management of water for irrigation, flood control, shipping, and recreation. In addition to the ESA, there is a long history of large scale management efforts to preserve Pacific salmon including the 1938 Mitchell Act, the Bonneville Power Fish and Wildlife Program (established under the Pacific Northwest Electric Power Planning and Conservation Act of 1980 Northwest Power Act 16 USC § 839b(h)(10)(A)) and the 2008 Columbia Basin Fish Accords - Bonneville Power Administration (Northwest Power and Conservation Council (NPCC), 2021).

The influence of Pacific salmon as a KMS continues south into California where commercial and recreational fisheries for salmon have generated between \$23.5 million and \$82.9 million in personal income in recent years (Pacific Fishery Management Council, 2024), making up roughly 0.001% of California's \$3.7 trillion economy. Rather than the value of the fishery, the real driver may be that the recovery of ESA listed Sacramento winter run Chinook and steelhead in California's Central Valley and other harvested stocks plays a critical role in the decision-making process to preserve ecosystem processes while stabilizing the fundamental infrastructure that provides freshwater to ~30 million people in San Francisco, Silicon Valley, and Los Angeles, along with irrigating ~\$66 billion in agricultural products and related economic activity which provide about 13% of the US agricultural production by value (United States Department of Agriculture (USDA) Farm Service Agency, 2011).

Coast wide, the ESA has played a strong role in local and watershed management decisions, driving a coexistence approach between serving the fundamental needs for food, water, and energy for more than 54 million people (worldpopulationreview.com accessed 5/21/2023) while supporting or preserving at least part of a functional ecosystem for the target species thus providing umbrella protections extending to all the other species and habitats they require to survive. These protections extend to 40% of the 1.05 million km² of area represented by 4 states (Figure 3). While declines in commercial fishing can result in reduced harvest pressure under the Magnuson–Stevens Fishery Conservation and Management Act (MSA), it is the ESA that really drives the habitat restoration for salmon habitat that benefits both listed and harvested stocks. The benefits to unlisted stocks spread to



commercial enterprises, such as fisheries and food services, as well as indigenous communities that depend upon salmon for food and cultural needs (Atlas et al., 2020; Norgaard, 2019; Reid et al., 2020). The Pacific salmon ESA influence on land management practice extends to forestry, agriculture, urban development, private and public land conservation and beyond. This benefits everything from the obscure endangered Ohlone Tiger Beetle (*Cicindela ohlone*), endemic to the recovered redwood (*Sequoia sempervirens*) forests of Santa Cruz County, to coast wide populations of mountain lions (*Puma concolor*). Finally, there have been many other unexpected benefits from salmon management. The need to track the survival of a fingerling Chinook smolt through a hydropower dam to evaluate passage

standards has necessitated the development of miniaturized tracking technology. Much of the miniaturized telemetry technologies ranging from radio tags to the microchips now used globally in pets, livestock and wildlife research were developed to meet the needs of Pacific salmon science and management on the Columbia River (Prentice et al., 1986).

Despite unprecedented investments, the future of many endangered Pacific salmon stocks remains precarious (Jaeger and Scheuerell, 2023). While 1 or 2 stocks could potentially be considered for down- or delisting, climate change is exacerbating many of the current management challenges. A general rethinking may be required around what stock recovery means in the face of climate change (Kocik et al., 2022). Regardless of approach, it is likely that Pacific salmon will remain a top driver of ecosystem-based management along the West Coast.

2.3 North Atlantic right whale *Eubalaena glacialis*

Large whales influence ocean ecosystems in numerous ways, including moving nutrients vertically and horizontally, in processes described as the whale pump and great whale conveyor belt, and creating whale-fall communities when they die and sink to the deep sea (Roman et al., 2014). Many whale species and populations were depleted by commercial whaling and other human activities, reducing their role as nutrient vectors and marine ecosystem engineers (Roman and McCarthy, 2010; Savoca et al., 2021). Despite the end of whaling, many of these species now make different and significant contributions to the economy. They have a charismatic appeal to humans as demonstrated by the hundreds of nature documentaries and the thousands of whale watch trips each year. All of these social and ecological values have been codified into extra protections of these species through the Marine Mammal Protection Act. Given the charisma of these species, one might expect humpback whales (*Megaptera novaeangliae*) to be a KMS with a high level of investment given their tendency for aerial acrobatics and beautiful songs. ESA investments for this species's recovery across multiple stocks was often ranked in the 'top 100' for species with highest level of investment as reported by USFWS, peaking just under \$8 million in 2010. Their recovery in the post-whaling era has been quite successful with 9 of the 14 distinct population segments (DPS) worldwide being delisted, including the western Atlantic Gulf of Maine population (Bettridge et al., 2015) and current recovery funding was last reported at \$1.7 million across 3 DPS's in 2020.

Rather, it is a more obscure and perhaps less charismatic species that has emerged as the management focus of the western Atlantic. The North Atlantic right whales' diminished population size and endangered status makes them a deciding factor for the co-management of nearly every marine economic activity in their range, driving the recent funding increase in management solutions (Figure 2). Specifically, NMFS must coordinate and apportion risk to North Atlantic right whales across the entire western Atlantic including the shipping industry and food and energy sectors.

North Atlantic right whales (hereafter right whales) once occurred across the North Atlantic in numbers between 10,000 and 20,000 (Monsarrat et al., 2016), in one or more populations (Frasier et al., 2022). The harvest of this species occurred for centuries in Europe and continued in North America until initial protections were put in place by the International Convention for the Regulation of Whaling in the 1930s (Reeves and Smith, 2006). Current stock assessment reports describe a remnant population residing primarily along the east coast of North America with a stock estimate of roughly 270 animals in 1990. This rose to almost 500 animals by 2011, followed by a rapid decline to just under 360 animals in 2020 (Linden, 2024). This recent decline began with animals changing their movement patterns in search of prey resources that shifted in response to climate change and accelerated warming in the Gulf of Maine. The whales moved into areas with less management protection from long standing threats, increasing their exposure to vessel strike and fishing-gear entanglement (Meyer-Gutbrod et al., 2021, 2023). The reduced population leaves no resilience for added threats, thus triggering significant management activity in two sectors.

- To reduce vessel strikes, multiple actions have been taken that required significant coordination with the shipping industry. Historically this has focused on reducing the probability of encounters, including moving the shipping lanes into key harbors to areas with fewer right whale sightings ((CFR), C.o.F.R., 2007, Federal Register, 2010) and reducing vessel speed to give whales and mariners more time to react and avoid strikes (Conn and Silber, 2013, Federal Register, 2008). Automated Identification Systems (AIS), a vessel-based tracking system that broadcasts ships' locations, are used both for enforcement and acknowledgment of 'good behavior' as well as for modeling the probability and risk of vessel strikes under varying vessel speed and size classes and whale densities (Blondin et al., 2025). Research is underway for the use of advanced technologies like thermal imaging to increase the detection probability of whales (specifically their warm air blows) by transiting ships (Zitterbart et al., 2020). NMFS continues to pursue combinations of these solutions to reduce vessel strikes to right whales and other large cetacean species to manage coexistence with all aspects of the maritime industry along the eastern seaboard.

To reduce fisheries entanglement, managers have increased the use of traditional methods such as fishing closures and reducing the number of vertical lines by trawling up, the act of stringing more pot traps together at depth. Managers have also worked with fishermen to reduce the breaking strength of vertical lines, so animals have a better chance of escaping entanglement when it happens (Knowlton et al., 2016). But the most novel solution has been the development of ropeless or on-demand fishing technologies that remove the vertical line and buoy which traditionally mark the position of trap pots and gillnets on the seafloor, replacing them with a technologies that send a line-connected float to the surface, only after a fisherman has arrived to recover the gear (Baumgartner et al., 2019; Matzen et al., 2025). This enables fixed gear fisheries to continue harvesting in high-risk areas by reducing or eliminating animal exposure to the main source of entanglement. A key outcome of this is the development of near real-time data systems

to inform ocean users of the presence of gear on the seafloor. The loss of the centuries-old communication system of the buoy necessitates the development of seafloor traffic applications analogous to those used for GPS navigation with smartphones, and it is possible this data system will eventually extend across many ocean-use sectors.

At the same time, new challenges are emerging for right whales. Construction has begun for thirty gigawatts of offshore wind farms to be potentially completed in the mid-Atlantic by 2030 with more by 2050 expanding into the Gulf of Maine, along with coast wide increases in off-shore aquaculture for protein production and kelp farming (food, carbon capture and biofuel) also being considered. These have the potential to bring new risks including disrupted oceanographic patterns affecting forage, increased noise exposure and attraction of predators, while exacerbating old threats including increased vessel traffic and entanglement (BOEM and NOAA, 2024).

The management practices focused on right whales will have extended benefits to many taxa and much of the ecosystem they inhabit. The large-scale reduction of vessel speeds or avoidance technologies will benefit many large vertebrates affected by the same threat of vessel strike, from cetaceans to sharks and turtles. Similarly, fishery management practices worldwide will benefit from a significant investment in on-demand fishing, an expensive technology that is heavily subsidized by the ESA investments reported here. It is likely that this technology will revolutionize fishing practices, both through the provision of big-data opportunities to track stock distributions driven by climate change and the reduction of fishery bycatch at a global scale. The georeferenced data system also has the potential to emerge as the basis for a global ocean maps system that can track other activities on the seafloor for marine spatial planning.

Perhaps one of the more recent phenomena in the human dimensions of charismatic KMS like North Atlantic right whales is the increasing focus on these species by social media. While often viewed as a tool to increase awareness for the conservation cause, KMS are also proving vulnerable to the darker side of social media including the spreading of misinformation to promote a different agenda. An example of this was documented in a *Science Friday* news story that chronicled a social media narrative around a right whale mortality caused by entanglement that evolved to blame offshore wind development activities (Duhaime-Ross, 2024). The story documents how the social value of several large whale species including right whales are being leveraged by both sides of national and likely global debates around renewable energy expansion for which there is already an interesting relationship between conspiracy beliefs and opposition (Winter et al., 2022). This is a complex topic with tremendous uncertainty and highlights a point of caution in the emerging concern over scientific misinformation, that the cause or value of these high profile species can be hijacked for ulterior economic and political motives. The advent of this has created a complicated and rapidly changing environment that increases the challenges of wildlife recovery and ecosystem-based management.

2.4 Grizzly bear *Ursus arctos horribilis*:

“The government trapper who took the grizzly knew he had made Escudilla safe for cows. He did not know he had toppled the spire off an edifice a-building since the morning stars sang together.... Escudilla still hangs on the horizon, but when you see it you no longer think of bear. It’s only a mountain now.” -Leopold 1966

The grizzly bear is the climax predator of North America, yet unlike strict carnivores such as wolves and lions, the bear is an omnivore with a diet consisting as much on berries, pine seeds and nuts, insects, rodents, and salmon and trout as the game and livestock it is capable of hunting or scavenging (Gunther et al., 2014, USFWS 1993b). This diet has likely given it a complex relationship with humans: it is both feared as a “man-killer” and affectionately caricatured as the picnic-basket stealing Yogi Bear of Jellystone, a likely follow on to the bear-viewing practices in the garbage dumps of Yellowstone National Park (~1890-1940, Dickson, 2023).

Historical populations were estimated at 50,000 in the Lower 48 states with a range from California to the Great Plains and south into Mexico. By the time of their ESA listing in 1975, grizzly bears in the Lower 48 were reduced to 700–800 animals in 2% of their range (United States Fish and Wildlife Service (USFWS), 2022). Their decline is attributed to multiple forms of human-caused mortality, overharvest of prey species such as bison and elk, and habitat loss. Recovery has come about through habitat restoration and reduced fragmentation, strategies to minimize human-caused mortality, recovery and management of large-game herds and public education. Grizzly bear populations in the Lower 48 states have significantly expanded since the time of listing in 1975 and now occupy approximately 6 percent of their historical range in the Lower 48 States, and number over 2, 000 individuals in 4 ecosystems. In the Greater Yellowstone and Northern Continental Divide ecosystems, populations are now at or approaching recovery. The ESA status of the species has been debated in litigation in a variety of settings for most of the last fifteen years (Greenwald, 2023) and continues today as noted by 3 recent petitions to delist (Federal Register 2023).

The grizzly bear in many ways appears as a classic umbrella species. Its preservation drives land-use practices and supports the large-scale preservation of federal lands and wilderness areas. However it’s a bit more complicated. Protections began with the formation of national parks and forests, Yellowstone being the first in 1872, then they were codified in the Wilderness Act of 1964, creating strongholds for this species. The ESA listing in 1975, gave the bear actual protections beyond these areas, extending anywhere they may be present. This has created the challenge of making the human landscape permeable to bears, forcing agencies to address issues like roads, development and unsecured attractants leading to a greater coexistence approach. In addition, grizzly bears’ effect on EBM is seen both through federal regulations and litigation around

issues of logging, roads, mining, oil and gas leasing, livestock grazing, and recreation (Greenwald, 2023).

Currently there are six designated grizzly bear recovery areas covering 100,673 km² much of it national parks, wilderness areas and federal lands, plus an additional 42,575 km² of buffer zone (United States Fish and Wildlife Service (USFWS), 2022). Actions in these zones have included retiring grazing allotments and augmentation programs. In addition, conservation organizations are buying and protecting land related to these recovery areas with contributions in excess of 2,700 km². It is these areas where their KMS effect expands. This habitat preservation has umbrella benefits for other species including game, carnivores and greater forest ecosystems. As the USFWS is considering delisting, the states and land managers have drafted conservation strategies that are meant to demonstrate “adequate regulatory mechanisms” that will remain in place post delisting for a species that will likely require everlasting management strategies to address conflict mitigation with humans (Pers comm D. Diamond).

Grizzly bear investment has averaged \$7.2 million/year since 1989 (Figure 2). It has had the most stable investment trend among the selected examples, increasing slowly over time, with some pulses for investments in land acquisition. Otherwise, most funding is dedicated to research and monitoring, infrastructure, conflict reduction, and education. The grizzly has the smallest current home range of our examples (at least the population in the Lower 48 states), but it has remained in the top tier of funding, initially in the top 10 highest investments. Why invest in grizzly recovery? In short, people feel a strong connection to this animal. They seem to be as fascinated with them as other predators like the great white shark (*Carcharodon carcharias*). In Montana, the grizzly bear is the state animal, the logo of the state wildlife agency, and the mascot of the flagship state university. A recent study reported that 75% of Montanans agreed that it is important to maintain a self-sustaining grizzly bear population (Costello et al., 2020). The story represents the most persistent desire for literal coexistence. This is an animal that lives on the ground with the potential to walk among us. With that comes a fear for human safety and threat to individual livestock, but these are issues that strike more of an emotional issue around what become very public events. Relative to the mundane more frequent risks of car crashes and livestock getting sick, the frequency of grizzly bear conflict events are quite low. While this can be achieved through some greater separation between animals and population centers and removal of conflict animals, it can also be the result of successful education and people making life choices to manage refuse, gardens and other food attractions that might otherwise attract animals and create conflict.

Human development has continued with significant pressure for increased residential (93,000 homes built 1990–2018) & commercial development in grizzly bear occupied areas, ironically because there is so much wilderness recreational opportunity (Nesbitt et al., 2023). While livestock rearing and predators are viewed as a social conflict, subsidies are paid through the United States Department of Agriculture (USDA) Farm Service (2025) for any depredation. This serves to offset the economic impacts of coexistence even if it doesn't resolve more emotional ones. The recovery has occurred during people's lifetimes.

They remember ranching without bears, and now they have bears. And as the range expands, more ranchers are getting to experience it.

Also because of human safety risks occurring with regular encounters to habituated large predators, significant resources and education are necessary to reform residential food storage and waste management practices to ensure ‘wild’ bear behaviors that are naturally predisposed to avoid humans. This is part of why we considered grizzly bears to be more than an umbrella species provided with rural habitats that also benefit other species. The jump to being a KMS involves the clear (and for some, surprising) intent to have urban areas permeable to grizzly bears. The challenge is that these programs will likely need to be maintained into perpetuity and expanded as more humans move into the region and bear populations increase (Roach, 2021). If delisted, their management would transition to state game offices and there is the potential for hunting to be reauthorized. They also have several habitat protections including the designation of Yellowstone and Glacier National Parks and the Wilderness Act. Grizzly bears have a long history of conflict with ranching, but perhaps this is evolving into one of symbiosis, where the two are intertwined in the need for land preservation. While Montana's human population remains relatively small, it has grown consistently throughout its history, at roughly 0.9%/year for the past 25 years (commerce.mt.gov) and with this a continuous increase in homebuilding. It may be that long-term connectivity goals will require working lands as part of a permeable landscape between recovery zones (which are mostly public parks and wilderness land).

There is clear expansion of grizzly bear occupied range, and the emerging articulation of a connectivity goal by resource agencies. There are also resources flowing to working lands from sources like the Bipartisan Infrastructure Law for conflict reduction related to grizzly bears and wolves. The public lands are well managed, so private land conservation is the future for a host of conservation interests, and keeping working lands working and not converted is critical. Grizzly bear connectivity is one way that helps people grasp the importance of this issue. This in turn has a significant feedback loop on land management and agricultural practices.

3 Discussion

3.1 Why do KMS occur?

Many species qualify as umbrella or flagship species, which can serve to justify management actions such as setting aside vulnerable habitat to the exclusion of some human activity. For some species, habitat requirements overlap with human uses, creating conflict and a need for coexistence, thus driving EBM of our own actions while ensuring that the ecosystem requirements of a particular species is met, even if they're not restored to natural conditions. Given the sheer loss of biodiversity and dominance of humans on the global landscape (Bar-On et al., 2018), management actions will need to be increasingly focused on directing habitat use and management actions toward coexistence rather than resisting human presence and attempting the restoration to some preindustrial ecological state (Thompson et al., 2021). It is under these conditions where

keystone management species emerge, and it is likely management actions will be increasingly focused on this approach.

This potential for KMS to emerge increases when an endangered charismatic species comes into conflict with some human resource need or has economic impact on a significant number of stakeholders. This has a maximal effect when the species conflict is with fundamental societal needs. One may postulate that when this conflict escalates to a political level, gaining the attention of elected officials in the US Congress, a KMS is 'born' in the United States. Although we focus on the Endangered Species Act in the United States, we suspect this is also an international phenomenon as in the case of the North Atlantic right whales, whose range encompasses most of the densely populated eastern seaboard and expanded into Canadian waters in the previous decade. Regulations affecting coastal vessel traffic, commercial fishing, offshore energy development, and aquaculture subsequently influence the coastal economies of 14 states (and 28 senators) from Florida to Maine and into the Canadian Maritimes. This species is frequently seen as a polarizing conservation issue, with stakeholder pressure to see the issues resolved, resulting in significant investments in science and coexistence solutions (Figure 2). In 2023, North Atlantic right whale investments in the U.S. reached the highest level ever invested in a single year for a mammal.

A slightly different scenario occurs for Pacific salmon in the contiguous United States. There are only 4 West Coast states (CA, WA, OR and ID) with anadromous salmon habitat represented by only 8 senators, yet there are 70 representatives (WA-10, Oregon 6, ID 2 and CA 52). Management decisions on these fish in this region directly influence food, water and energy security decisions supporting 19% of the US GDP (United State Bureau of Economic Analysis (BEA), 2023). In addition, while some species like right whales tend to be in a polarized political conflict between conservation and economics, Pacific salmon recovery has both direct conservation and economic benefits. Watersheds with commercially and recreationally harvested salmon stocks benefit from protections on co-occurring ESA listed stocks, producing

more fish for harvest increasing economic returns. Specifically, the influence of salmon and resources dedicated to their recovery are highest in watersheds and coastal regions with both ESA-managed salmon stocks as well as harvested salmon stocks managed under the MSA (USFWS ESA reports). This tends to be the larger watersheds, which biologically support Chinook salmon (the highest value salmon fishery for the region) and more people and their needs. Having an endangered stock brings the regulatory authority of ESA, but the umbrella effect extends benefits to harvested salmon stocks which are likely to benefit by things like improved fish passage and habitat restorations. Thus "salmon" becomes a unifying issue—both conservationists and harvesters/consumers want more salmon and it's often a winning political issue to support salmon recovery but serious conflicts tend to remain around water for agriculture.

In the 4 examples chosen we observed that each species has an additional regulatory component associated with the species. In the case of bald eagles and North Atlantic right whales- they have additional conservation protections under other laws including the Marine Mammal Protection Act, the Migratory Bird Treaty Act and the Bald and Golden Eagle Protection Act. In the case of salmon and bears, they have potential harvest and game value and are otherwise managed respectively as a renewable resource under the MSA as well as multiple state fish and game rules across their range should they be delisted.

It is also worth noting that all examples are vertebrates, a significant investment bias towards this taxonomic group (previously observed by Haines et al., 2021). With few exceptions, the top 50 species ranked by investments each year (Table 2 in the USFWS ESA reports) are vertebrates. White abalone (*Haliotis sorenseni*) has ranked that high in the most recent 2020 & 2021 reports, and slender rush-pea (*Hoffmannseggia tenella*) appeared once (1999 report). The vast majority of investment goes toward fish, dominated by salmonids and sturgeons, followed by birds then mammals. Figure 4 shows a typical annual investment distribution

Taxonomic breakdown of ESA investments for 2017

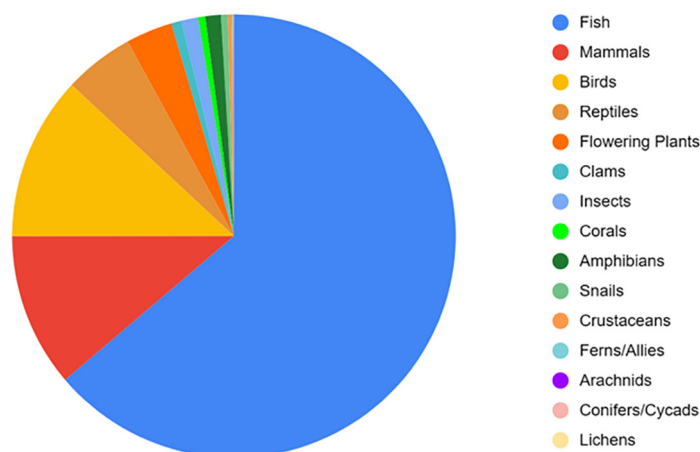


FIGURE 4

Taxonomic breakdown of ESA investments (\$1.256 billion) for 2017. (Does not include reported land acquisitions and multispecies investments.).

by taxa. The largest single year investment for any species may have been for the red-cockaded woodpecker (*Picoides borealis*) in 1992 for a Florida land acquisition valued at \$73.6 million (~\$168 million in 2024 dollars). Other species which have had regular spots on the “top 10 list” over the years include desert Tortoise (*Gopherus agassizii*), pallid sturgeon (*Scaphirhynchus albus*), Steller sea-lions (*Eumetopias jubatus*), northern spotted owls (*Strix occidentalis caurina*) and bull trout (*Salvelinus confluentus*) with annual investments ranging between approximately \$20–50 million.

Perhaps the defining reason might be that once the survival of a highly valued species comes into conflict with an issue of economic or strategic importance, it is possible that a species’ value or recovery investment scales to the value of the issue at hand: be it water, energy, health and safety, or food security. This likely explains the investment in Pacific salmon, where single stocks of steelhead and Chinook regularly exceed \$60 million per year (and as high as \$124 million). Similarly, the recent regulatory influence of right whales off the East Coast of the United States on activities including fishing, vessel traffic, and offshore wind development explains the sudden increased investment that rivals Pacific salmon stocks.

Finally, there is the issue of icons. At the risk of somewhat circular logic, these species are often symbols of a nation or region, often on flags, the likely derivation of ‘flagship species’. At least 6 countries have an eagle prominently on their flag, and while not on the American flag, the bald eagle is the national animal and the national seal. Similarly bears are the national animal of Russia and a common icon on flags. Interestingly, there seem to be few examples of any whale or salmon on national or regional flags, but both are common icons and art subjects in their region, particularly for salmon in the Pacific Northwest and whales in New England. The nature of the iconography may also be influenced by the terrestrial versus aquatic nature of species, one element being we are more aware of terrestrial species that are predators and thus viewed for their strength (e.g. lions are another common national flag icon).

3.2 Terrestrial versus aquatic species

Are KMS as likely to occur in all environments and to the same degree of influence? Our examples span multiple habitat types including terrestrial, freshwater and marine. We speculate that three key habitat variables may influence the nature and scale of human interactions with these high profile species. These include the issues of 1) private land ownership versus public commons, 2) species range and 3) the type and number of uses humans have for that habitat.

In the U.S., one can buy land, but not a river or the sea. Terrestrial species ranges on land can cross both public and private ownership. But rivers and oceans represent a commons where many humans and industries compete to extract financial gain from the water within the river or the same part of the sea, where there is no eminent domain. This has led to different EBM approaches to managing coexistence between endangered and often high profile KMS species.

The protection of terrestrial species can have large economic impacts, but recovery investments seem to be smaller. Large scale spatial agreements, such as Working Lands for Wildlife and Safe Harbor (now Conservation Benefits Agreements) have been reached for species like the lesser Prairie-Chicken (*Tympanuchus pallidicinctus*), the Greater Sage-Grouse (*Centrocercus urophasianus*), and the Red Cockaded woodpecker (United States Department of Agriculture (USDA), 2024, United States Fish and Wildlife Service (USFWS), 2017b, Federal Register 2024). These agreements were designed to give some economic certainty to land-users and developers as sufficient portions of habitat are set aside, or utilized more sustainably for protection of those focus species and their associated ecosystem members. At the same time these individual landowners are negotiating with managers to incorporate the recovery of endangered species into their own activities. For aquatic species, the Pacific salmon example seems to have the greatest economic and ecological impact, likely because it affects management decisions in marine, riverine, and terrestrial environments and human needs such as fresh water, food, and energy. Right whales are a purely marine species and their influence is felt across the commercial fishing, marine shipping, and offshore energy and aquaculture sectors.

Is a species more likely to be a KMS if it exists over a large versus small range? There is the element of the differing spatial scales at which terrestrial and marine animals exist depending upon whether they walk, swim or fly. The examples selected all have relatively large ranges relative to some species endemic to a small area. Right whale habitat use is basically the eastern continental shelf of North America from eastern Florida to the northern Canadian Maritimes region and likely beyond, so while different regions may use the ocean differently, the whale swims through all of them thus affecting all marine uses. Similarly, salmon move through all marine and watershed habitats crossing through multiple river and land use practices and economic activities as they migrate from the ocean to their spawning grounds. In contrast, grizzly bears are perhaps the most regional, but that is partly an artifact of their current recovery zone. Full reintroduction from the Rockies to the coast of California would be a more complex management issue. So the mechanism of migration (or size of home range) seems to be an element as well. Species that can fly over broken habitats between summer and winter habitats may be less influential in some respects. Or their influence grows wherever they touch down? The ending of DDT usage at a national and even global scale to preserve birds like the Bald Eagle is a key example. In contrast, perhaps a more endemic species like the snail darter (*Percina tanasi*), while a high profile conservation species might not have much of an EBM or keystone effect on a region, even if it had national legal impact (Ghezelayagh et al., 2025)?

There may be an economic scaling impact that comes with the single versus multidimensional use of habitats. Terrestrial lands tend to have one use e.g. forestry or agriculture, or are developed for commercial or residential use. Although that is changing, an example being the introduction of dual use photovoltaics and other land practices including agrivoltaics (Barron-Gafford et al., 2019). That aside, rivers and their freshwater resource, while in

some respects being ‘one-dimensional lines on a map’ are often multisector use of the water itself, perhaps because all the sectors in a watershed are influenced by how the water is trapped or free to move, either being used/stored for hydro-power, but also transported across the landscape for municipal and agricultural needs and serving as transportation lanes for commercial shipping of everything from grains and goods to petroleum. Rivers and their associated harbors attract large cities and industries like Seattle and Silicon Valley which are surrounded by and part of important salmon habitat. In further contrast, the 3-dimensional nature of the ocean allows for dual or multi sector use of any point of the map. While people do not live there, commercial fishing can harvest the bottom and water column, while shipping moves across the surface, and the emerging sector of offshore wind harvests energy above the surface. The protections for right whales influence all these sectors.

3.3 Are KMS a shield for the ecosystem?

The umbrella aspect of a keystone management species status is that they act as a regulatory shield creating cascading protections and recovery potential for most of the species and habitats that are part of that species’ food web and ecosystem. Pacific salmon act as integrators of ecosystems, since they depend upon all aspects of their watershed and marine ecosystems to be functional. Moratoriums were put on Pacific High Seas driftnet fisheries in the early 1990s, for example, in part to protect Pacific salmon (Huppert and Mittleman, 1993). Salmon also drive protections and restoration in the lower watersheds and estuaries of some of the most expensive real-estate in the world, including San Francisco, Seattle, and Portland. This moves up the watershed to agricultural lands that provide a large portion of the U.S. food supply and into mountains and headwaters where forestry practices are regulated by salmon-habitat needs. The resulting protections have cascading benefits to hundreds if not thousands of other species, including other less charismatic but also endangered ones, which may help their recovery or prevent ESA listings for other taxa.

Can there be intentional creation of a keystone management species? Similarly, Jarić et al. (2024) call for careful and strategic implementation of a flagship designation and even make the case for strategic use of what amounts to flagship individuals within a species. Examples include General Sherman, a giant sequoia believed to be the largest single-stem tree, and Sam the koala, rescued from Australian brushfires. Flagship individuals can affect social media with huge benefit, but it is a process that policy makers under the ESA may have little influence over, especially when that species or individual animal’s story is used by stakeholders for competing and conflicting agendas. Another recent analysis found some high-profile umbrella species have poor linkages to the biodiversity of the regions they inhabit, calling for an alternative approach—umbrella fleets (a multispecies approach) to include species overlooked by conservation decision-making (Ardiantiono et al., 2024). Similarly, Hazen et al. (2024) identified criteria for selecting appropriate species to monitor as ecosystem sentinels, serving an alternative purpose as metaphorical “canary” species

sensitive to ecosystem changes providing managers early warning signals. Some KMS might meet sentinel criteria, but in some cases they are likely to compete for management resources. While these ideas have merit, they may be difficult to implement given the reactionary nature of stakeholders and social media, which affects the focus and likely the investment in the US. This also represents an inherent conflict between data-driven scientific analysis and emotional response to inform what management priorities should be. In the case of KMS, science and management probably cannot and perhaps should not control which species receive this level of focus. But managers should be strategically aware of the issue and consider that in their approach to ensure the effectiveness of the ESA, such efforts can also protect ecosystem functions, as certain species’ influence scales with human conflict.

Would intentionally designating or creating keystone management species as a policy strategy be effective? Bald eagles have been delisted and recovered. Grizzly bears are under consideration for delisting. In contrast the billions spent on Pacific salmon recovery have been argued to have minimal effect in some cases (Jaeger and Scheuerell, 2023) although the EBM benefits are not fully appreciated. And it’s too early to tell if the recent investments in right whales will succeed. Investment in a KMS’s recovery continues only so long as it remains vulnerable. When recovery is successful, that investment goes away, but so does the need, and in the case of eagles, a new stable state is achieved for the ecosystem with many beneficiaries. It may be that current investments in Pacific salmon are just enough to prevent extinction in the face of forces like human population growth/development and climate change. Alternatively, the fact that so much is already invested to maintain status quo in some cases, suggests that new strategies and approaches are needed to truly achieve recovery and support coexistence, especially as these challenges are expected to increase.

3.4 A biased shield ... it’s still single species management

Under the guidelines of the ESA, which prohibits moving species into captivity to avoid extinction without plans to return them to the wild, you cannot save species without saving their habitat. This issue has been long debated for terrestrial and marine species conservation. A key argument against the idea of single species focus is whether it is enough to fend off the underlying problem—often human population growth, climate change etc. At times, management solutions tailored to single species may not be sufficient to benefit other species. Fish passage requirements through hydropower facilities, for example, are heavily driven by salmon swimming abilities and designs can fail to incorporate the needs of other less athletic species such as sturgeon, river herring, or lamprey especially if they are not also endangered. At the same time, one can argue that the massive investment in Pacific salmon conservation ensures protection of almost every watershed with a coastal connection along much of the West Coast (Figure 4), helping to conserve habitats from coastal redwood and alpine

forests to upland drainages, floodplains, and estuaries and without such a high-profile species, these protections might not exist.

Right whales, which suffer most of their mortality from entanglement and vessel strikes, have benefited from recent species-specific regulatory actions ([Federal Register 2021 FR 51970 Sept 17, 2021](#)) to reduce the number of ropes in the water and replace some of the strong vertical lines used for fishing-gear with weaker rope ([Knowlton et al., 2016](#)). The use of weaker ropes may have benefited some other large whale species, like humpback and fin whales (*Balaenoptera physalus*), but has likely done little for smaller species that experience regular entanglement like minke whales (*Balaenoptera acutorostrata*), basking sharks (*Cetorhinus maximus*) and leatherback sea turtles (*Dermochelys coriacea*). If new measures are implemented, they are likely to use closures to fishing with gear using vertical lines, but may allow access to fishing grounds with new technology, such as on-demand (aka “ropeless”) gear to prevent entanglement. This technology shift would benefit more species—at least in the regions it was used.

Finally, in the debate on whether to focus on saving an animal or its habitat, one must consider if there is a bias if the KMS is only dependent upon protecting a few elements in a particular habitat, leaving other elements vulnerable to impact with consequences to other taxa or as in the examples above, the management solutions only benefit that species. In the end it is usually the social/legal power of KMS that are driving the development of solutions and protections which will ultimately benefit those other species that would otherwise experience the same lack of investment as so many other listed species.

3.5 Keystone management species and ecosystem-based management?

While there are always clear elements of single-species management with KMS, in general the principles of EBM are found throughout the management approaches taken in recovering endangered species under the ESA. As mentioned previously, [Hare et al. \(2019\)](#) identified that the ESA practices of Atlantic salmon recovery mirror the approach of ecosystem-based fisheries management. However, a deeper linkage also exists. Much of the foundational literature of Integrated Ecosystem Assessment (IEA, e.g. [Levin et al., 2009](#); [Harvey et al., 2016](#)), which is NMFS’s defined approach for pursuing EBM, focuses heavily on recovery of Pacific salmon and their associated ecosystems. Most federal funding to NMFS related to Pacific salmon is for ESA recovery. In essence, the ESA investments described above likely drove the development of IEA principles partly in recognition that a different approach was needed to address the complexity of Pacific salmon recovery.

As suggested by [Haugen et al. \(2024\)](#), these same principles appear in other independent examples. Many of the IEA principles have emerged in efforts to recover right whales. Section 118 of the Marine Mammal Protection Act (MMPA), titled “Taking of Marine Mammals Incidental to Commercial Fishing Operations”, defines a stakeholder driven process that activates when bycatch of a stock

listed as threatened or endangered under the ESA, or depleted under the MMPA exceeds a mortality threshold. [Borggaard et al. \(2017\)](#) identified four principles to managing this process with regard to large whale entanglement, including 1) engage stakeholders, 2) establish a management strategy, 3) use a variety of management tools, and 4) incorporate adaptive management. The details overlap a great deal with the 5 elements identified by [Levin et al. \(2009\)](#) including 1) scoping, 2) indicator development, 3) risk analysis, 4) management strategy and evaluation, and 5) monitoring and evaluation ([Harvey et al., 2016](#)).

The MMPA process described by Borggaard et al., follows the IEA loop at every step:

- Scoping includes the formation of the stakeholder team, (typically composed of fishing industry members, academic experts, and conservation groups) when the target (whale mortality) is exceeded as well as public hearings when any regulatory action is planned. In the right whale example, NMFS formed the Atlantic Large Whale Take Reduction Plan and Team (ALWTRP/T) which is focused on interactions between multiple whale species right, humpback and fin whales and all fixed gear fisheries (trap, pot and gillnet) engaged in the harvest of multiple species of fish and crustaceans along most of the U.S. Atlantic coast.
- Indicators include regularly updated stock abundance and mortality assessments as well as increasing focus on ecosystem dynamics like preferred forage species (e.g., calanoid copepods, [Ross et al., 2023](#)) and shifting habitat distributions ([Roberts et al., 2024](#)).
- Risk analysis and management strategy evaluation for right whales is done through a population viability analysis that considers changing right whale demographics relative to their primary threats (entanglement, vessel strike, noise, and ecosystem change, [Runge et al., 2024](#)) and a decision support tool that quantifies risk across the landscape ([Shank et al., 2025](#)). The tool can simulate proposed changes in fishing practice (e.g., closures, changes in gear use, and effort shifts) to evaluate how risk might change as a function of different management strategies and the PVA can project how changes in risk and associated mortality will affect the population through time.
- Monitoring and evaluation are done through rigorous survey efforts to quantify right whale distribution, abundance, and mortality on an annual basis ([Linden, 2024](#)), and every time mortality exceeds PBR the team is reconvened to repeat the process.

3.6 Next steps

While our intent is to avoid the direct prescription of new policy, the observations and criteria defined for keystone management species necessitate a thorough discussion and raise

multiple topics for consideration by the scientific, management, and policy communities.

Our initial and perhaps most fundamental objective is to stimulate debate regarding the core concept itself. Is the KMS concept a truly unique, stand-alone construct, offering a novel lens through which to approach conservation and management? Does it occur in other countries with different approaches to endangered species and ecosystem management? Australia's Environment Protection and Biodiversity Conservation Act, for example, protects both threatened species and ecological communities, and as such, does not depend on individual species as KMS. Or, is the concept an operational extension of concepts already well-established in the ecological and conservation literature, such as umbrella, flagship, sentinel, or keystone species? A clear consensus on its distinctiveness is crucial for determining its utility.

Should the KMS concept be affirmed as a unique and valuable framework, several critical questions immediately emerge, demanding careful consideration:

- **Validation of Definitions and Criteria:** Are the definitions and criteria proposed for identifying a keystone management species accurate, sufficiently rigorous, and practically applicable in diverse ecological and jurisdictional contexts?
- **Policy Formalization and Statutory Integration:** Does the KMS concept warrant the establishment of formal policy criteria? While we expect KMS will almost always emerge in retrospect in response to some increasing conflict, is there a compelling argument for integrating this concept into existing legislative frameworks, such as the Endangered Species Act (ESA) or other national or international conservation mandates? Formal recognition could provide new tools and legitimacy for strategic conservation efforts, but requires careful evaluation of potential legal and regulatory ramifications.
- **Strategic Integration of Management Approaches:** Could the KMS concept serve as an effective mechanism to strategically integrate traditional single-species management (which often dominates conservation efforts) with broader ecosystem-based management (EBM) approaches? KMS could potentially bridge this gap by focusing intense recovery efforts on a single, vital species whose recovery yields disproportionately large, measurable benefits for the entire ecosystem.
- **Maintaining Scientific Integrity in the Public Sphere:** How can managers and policy makers effectively maintain scientific integrity and the evidence-based nature of conservation decisions when communicating about KMS, especially in the contemporary landscape dominated by social media and rapid, often misleading, misinformation? The high-profile nature of a KMS could make it a target for misrepresentation, underscoring the need for robust, proactive science communication strategies.
- **Risk/Benefit Analysis of Formal Recognition:** What are the comparative benefits and risks to management associated with a formal, strategic approach to identifying and managing KMS? A formal process offers clarity and

intentionality, but may also introduce unforeseen political liabilities or divert resources from other critical species.

- **Investment Strategy and Recovery Effectiveness:** Will the formal recognition of KMS significantly affect how conservation investments are made across overall recovery efforts? Furthermore, are strategies that specifically target investments towards KMS recovery demonstrably the most effective and cost-efficient approach for achieving broader biodiversity and ecosystem health goals? This requires modeling and empirical evaluation to ensure resources are maximized and not merely shifted without true strategic gain.

3.7 KMS as our conscience and guide?

It is very difficult to affect change in the looming ecosystem challenges of human population growth, consumption and climate change directly. The scope is so daunting that no single action can have a measurable local benefit without global collective action, representing a Prisoner's Dilemma challenge at every level from the individual to multinational institutions- each must choose to take necessary action, assuming most others will too. The charisma of KMS captures the public's attention in ways that perhaps it makes 'the problem' more manageable to the average individual or institution who may feel helpless to affect the greater issues. Rather than targeting 'fixing' climate change, we are promoting 'recovery' of iconic species. And because these species tend to have large home ranges combined with a network of such species at a global scale, there is the potential for a collective global influence.

It has become clear that science and logic are insufficient motivating factors for the global population and economy to act to address climate change. It may be that policy makers and scientists need to be strategic about understanding other forces that motivate people such as spiritual or religious factors, or connections to the land and earth. These keystone management species may provide some bridge between science and emotional context that serves this purpose. However, in this era of science denial and manipulation, there is significant risk that social media can be used to misrepresent issues. Specifically, it is likely that many necessary climate actions may have unavoidable negative consequences to KMS or other public resources at the local level. The examples about right whales provided above suggest that short term consequences to high profile species will be a key argument to avoid climate mitigation, even though long term impacts of climate change for those same species may be quite dire, creating very difficult management decisions.

The reality is that our conservation laws like the ESA and the MMPA are social constructs. These laws are seen as managing non-human species, but are really managing people to ensure persistence of those species and it becomes a social science issue in that the charisma of these species inspires humans to do something for the ecosystem that likely benefits humanity. Yet without these species, it seems unlikely when asked to just fix the ecosystem humanity will prioritize it. In many respects, the social value of keystone management species represents a parallel consideration to the emerging climate change

strategy focused on considering impacts to future generations (Graham and de Bell, 2021; Revesz and Shahabian, 2010; White, 2022). Like KMS, future generations of children cannot speak for themselves. Perhaps these species give voice to both their needs today.

Keystone management species may or may not directly influence the ecosystem itself, but they influence, even inspire, humans to do multisector management. Because the ESA is based on the principle of static ecosystems, our management of these species usually is focused on ecosystem restoration to remove some level of human impact. Climate change is making this less and less feasible, and how the ESA works to affect recovery in a changing climate is something that needs serious consideration (Robbins, 2015; Ruhl, 2008). Having celebrated the fiftieth anniversary of the ESA, it's time to consider if the next 50 years will be defined as a merged strategy between single species and ecosystem-based management of all these sectors. In essence, the ESA is a recognition of self-regulation, which is a group selectionist practice. Indigenous practices aside, many Western thinkers have put forward the idea that humans need to shift their view of being separate from an ecosystem that is a resource we compete to capitalize on to being one with it, for mutual benefit of humanity and the ecosystems on which we depend (Farley et al., 2024; Robbins, 2015; Leopold, 1966). A challenge with achieving that transition is that the average person in a global economy dependent on urbanization for efficiencies will never appreciate this complexity, nor have direct connections to 'nature'. Similar to the 'species-on-the-move' concept (Pecl et al., 2023), keystone management species may be the closest link many people can make to such principles and thus are the necessary bridge scientists and policy makers need to shift our institutional approach to resource management. Perhaps if the world one day achieves an economy no longer dependent upon continuous growth, its roots will be traced to conservation laws like the ESA. At a superficial level some might say that preservation of these species and the ESA makes no economic sense, but if these species are the guardrails of progress to ensure that some amount of the natural ecosystem is preserved no matter what, maybe it makes all the economic sense in the world.

Author contributions

SH: Formal analysis, Writing – review & editing, Writing – original draft, Conceptualization, Investigation. JR: Conceptualization, Writing – original draft, Writing – review & editing.

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Conflict of interest

The authors declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

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